Leveraging Machine Learning for Strategic People Analytics: Predicting Employee Attrition, Retention Strategies, and Predicting Employee Performance

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By Narjis Jebali Giacalone
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Thesis Mentor: Dr. Long Xia

Approved by (tentative):

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Name of Professor, Department (Thesis Mentor)

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Name of Professor, Department (Reader)

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Name of Professor, Department (Reader)

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Abstract

Employee attrition significantly impacts organizational efficiency and financial performance, necessitating effective predictive analytics to address this challenge proactively. Despite substantial research applying various predictive modeling techniques to attrition, existing literature often overlooks essential methodological considerations, limiting practical effectiveness. This study systematically evaluates recent predictive modeling approaches to employee attrition, identifies key methodological gaps, and introduces several advancements. Specifically, it implements state-of-the-art ensemble methods such as XGBoost (eXtreme Gradient Boosting) and LightGBM (Light Gradient Boosting Machine), integrates interpretability through explainable AI, assesses models for fairness and bias, and utilizes robust cross-validation strategies to enhance predictive accuracy and ethical validity. By addressing these methodological shortcomings, the study delivers a comprehensive predictive framework that supports strategic decision-making in managing employee attrition, retention, and performance.

Keywords: Employee attrition; retention strategies; employee performance; predictive analytics; ensemble methods; explainable AI

1. Introduction

Employee attrition—defined as the gradual reduction of workforce through voluntary resignations, retirements, or internal transfers—has become a critical issue in strategic human resource management (Society for Human Resource Management [SHRM], 2023). Unlike layoffs or involuntary turnover, voluntary attrition typically stems from employees' personal decisions to leave the organization, often due to factors such as inadequate compensation, work-life imbalance, misalignment with organizational culture, or limited career advancement opportunities. For the purposes of this study, attrition is defined specifically as voluntary attrition; involuntary separations such as layoffs or terminations are outside the scope of this research. Although some degree of attrition is natural and expected, elevated voluntary attrition rates—particularly in high-turnover sectors such as technology and services—can have serious consequences for organizational stability, productivity, and long-term competitiveness.

High attrition leads to substantial costs related to recruitment, training, and lost productivity, as well as erosion of institutional knowledge and cultural cohesion (Hom et al., 2017; SHRM, 2023). Moreover, chronic attrition can burden remaining employees, increasing burnout and leading to further turnover in a self-reinforcing cycle. Given these cascading effects, organizations increasingly seek robust, proactive tools to predict and mitigate attrition risks before they manifest operationally.

Predictive analytics—especially those enabled by advanced machine learning (ML) techniques—offer promising solutions for identifying employees at elevated risk of leaving, enabling targeted, data-driven retention interventions (Zhao et al., 2020). However, many existing approaches suffer from limitations including suboptimal model selection, poor interpretability, limited fairness analysis, and inadequate validation, undermining their practical value for managerial decision-making.

This study addresses these gaps by evaluating predictive analytics methods used for employee attrition, proposing methodological enhancements to increase their reliability, transparency, and ethical soundness. Specifically, it introduces advanced ensemble models such as XGBoost and LightGBM, integrates explainable AI (XAI) methods to improve interpretability, and applies fairness auditing to ensure equitable decision support. In addition, this research extends beyond prediction to include actionable retention strategies and preliminary exploration of employee performance forecasting as part of a comprehensive people analytics framework. The goal is to deliver a more effective, explainable, and ethically grounded predictive methodology to support strategic HR decision-making.

2. Literature Review

Given that this study examines the application of machine learning models to predict employee attrition and enhance strategic human resource decision-making, we review three directly relevant research streams: (1) predictive modeling techniques for employee attrition, (2) interpretability and fairness in predictive analytics, and (3) validation approaches in employee attrition studies.

2.1. Predictive Modeling of Employee Attrition

Employee attrition prediction has garnered considerable research interest, primarily leveraging publicly accessible datasets such as those provided by Kaggle and IBM HR Analytics. Recent studies employing these datasets have utilized various machine learning algorithms, including Random Forest, AdaBoost, and deep learning models (Arqawi et al., 2022; Zhao et al., 2018; Sisodia et al., 2017). Arqawi et al. (2022) highlighted the efficacy of deep learning approaches, achieving superior predictive performance (94.52% F1-score) compared to traditional machine learning models (around 92.55%). Zhao et al. (2018) and Sisodia et al. (2017) found Random Forest models frequently outperforming other algorithms such as Decision Trees, Support Vector Machines (SVM), and Naive Bayes in attrition prediction.

Common predictors identified in these studies include job satisfaction, monthly income, tenure, job involvement, and stock options. Although these variables consistently emerge as significant predictors, the predictive accuracy of traditional machine learning approaches typically reaches a ceiling, while deep learning models have incrementally enhanced overall prediction metrics (Arqawi et al., 2022).

2.2 Interpretability, Fairness, and Validation Challenges

Despite these advancements, the existing literature reveals critical limitations. Many studies rely predominantly on simpler classification methods or basic neural networks, overlooking advanced ensemble methods such as XGBoost and LightGBM, known for their superior performance, efficiency, and effectiveness in handling imbalanced datasets. Additionally, interpretability remains inadequately addressed, as current studies provide superficial explanations of model predictions, hindering practical applicability.

Another notable gap involves the absence of fairness and bias evaluations within predictive models. Given the ethical implications in human resource analytics, the failure to systematically evaluate model fairness presents a significant shortcoming. Moreover, existing studies often employ simplistic train-test splits, risking overfitting and poor model generalizability, highlighting the need for more robust validation methods such as cross-validation or stratified sampling.

2.3 Research Gaps and Our Study

Upon reviewing the extant literature, several key research gaps become apparent. Firstly, there is a marked underutilization of advanced ensemble machine learning models like XGBoost and LightGBM in attrition studies, despite their potential advantages in predictive performance and efficiency (Research Gap 1). Secondly, model interpretability has been superficially addressed, lacking in-depth exploration through explainable AI techniques, thereby limiting actionable insights for practitioners (Research Gap 2). Thirdly, existing literature rarely assesses fairness and bias within predictive models, neglecting the ethical dimensions crucial to HR analytics (Research Gap 3). Finally, methodological rigor is often compromised due to simplistic validation approaches, affecting the generalizability and practical reliability of predictions (Research Gap 4).

To bridge these gaps, our study introduces several innovative aspects. We employ advanced ensemble techniques, specifically XGBoost and LightGBM, to enhance predictive performance in attrition modeling. Additionally, we adopt explainable AI methods, such as SHapley Additive exPlanations (SHAP), to deliver comprehensive model interpretability and actionable insights. We also systematically evaluate model fairness using established frameworks such as IBM's AI Fairness 360 toolkit to ensure ethical compliance. Finally, we integrate robust validation techniques, including cross-validation and stratified sampling, ensuring the generalizability and reliability of our predictive models. Through these methodological enhancements, our study aims to deliver an advanced, interpretable, ethically sound predictive analytics framework capable of effectively supporting strategic HR management decisions.

3. Research Framework & Methodology

The primary objective of this research is to develop a robust, interpretable, and ethically sound predictive model to identify employees at risk of attrition. By accurately predicting which employees are likely to leave the company, HR management can intervene proactively, reduce turnover costs, and retain key talent. To achieve these goals, this study employs advanced ensemble machine learning models to enhance predictive accuracy. Additionally, interpretability of the predictive models is prioritized by utilizing SHapley Additive exPlanations (SHAP), ensuring that model predictions are transparent and actionable. Fairness assessment is systematically performed using IBM's AI Fairness 360 toolkit to ensure ethical model deployment. Furthermore, robust cross-validation methods, such as stratified K-fold cross-validation, are employed to enhance the reliability and generalizability of the predictive results.

The dataset utilized in this research is sourced from the Kaggle repository titled "IBM HR Analytics: Employee Turnover Analysis" last updated in January 2025 by Miracle 'Nifise. This dataset comprises comprehensive information on 1,470 employees across 35 distinct features, including demographic, job-related, and performance indicators. The target variable "Attrition" categorizes employees based on whether they have left ("Yes") or remained ("No") with the company, providing a clear outcome measure for predictive modeling.

The methodological framework encompasses several key phases. Initially, data preprocessing involves addressing missing values, encoding categorical variables using techniques such as one-hot encoding, and feature scaling to ensure optimal model performance. Additionally, to address the significant imbalance in the target variable "Attrition," Synthetic Minority Oversampling Technique (SMOTE) is employed to generate synthetic examples of the minority class (employees who left), ensuring balanced class representation and improving model generalization. Advanced ensemble algorithms, specifically XGBoost and LightGBM, are then applied due to their robustness, efficiency, and strong predictive performance, particularly suitable for handling imbalanced datasets common in attrition scenarios. Model validation incorporates rigorous cross-validation approaches to ensure model robustness and prevent overfitting. SHAP values are utilized extensively to interpret model predictions, clearly identifying key factors influencing employee attrition. Finally, the developed predictive models will be practically deployed using an easy-to-use web-based tool. The deployment strategy involves serializing the best-performing model into a pickle file and deploying it using a Google Cloud Platform (GCP) function. This GCP function is then leveraged in the back end by a simple react web app. This web app functions as an HR tool that will allow HR personnel to upload employee data easily and receive straightforward predictions on employee attrition risks, providing clear, actionable insights. This web app would help HR personnel to communicate more effectively with the company's workforce. If the attrition risk gets high for certain employees, HR departments could plan out meetings and develop retention strategies to save time and cost that would have been spent hiring a replacement employee or lost due to inactivity.

4. Predicting Employee Attrition

This section presents a detailed step-by-step approach to developing, evaluating, and deploying predictive models for employee attrition. Initially, an Exploratory Data Analysis (4.1) is conducted to understand data characteristics and feature relationships. Subsequently, the process of Model Development and Cross-Validation Results (4.2) is outlined, demonstrating the rigor of model training and selection. The effectiveness of the chosen model is further validated through a Hold-out Test Evaluation (4.3), ensuring its reliability. To enhance transparency and practical applicability, Explainability and Model Understanding (4.4) techniques are implemented. Finally, Deployment (4.5) strategies are discussed, focusing on practical applications of the predictive model within organizational settings.

4.1 Exploratory Data Analysis

This exploratory analysis begins with an examination of the IBM HR Analytics dataset, which includes 1,470 employee records and 35 features. Initially, all columns were assessed for missing values, revealing that the dataset is complete without any missing data. Several irrelevant columns were removed manually, including EmployeeCount, EmployeeNumber, MonthlyRate, Over18, and StandardHours, as they provided no meaningful predictive information for attrition. Analysis of the target variable "Attrition" demonstrated a significant class imbalance, with approximately 84% of employees not leaving ("No") and only about 16% leaving ("Yes"), highlighting the potential for biased predictions towards the majority class (Figure 1).

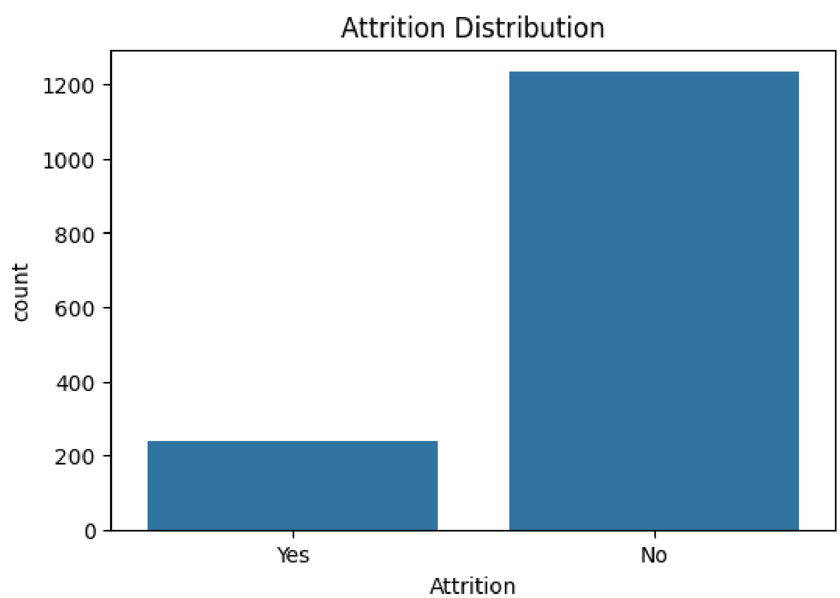

Figure 1: *Distribution of Employee Attrition (Original Imbalanced Dataset)*

Further analysis through descriptive statistics revealed insightful patterns: employees' average age was approximately 37 years, with a standard deviation of roughly 9 years, indicating a relatively wide age distribution ranging from young professionals to near-retirement workers. The average monthly income was approximately $6,500, with a significant standard deviation of about $4,700, highlighting substantial differences in employee compensation levels. DistanceFromHome averaged around 9 miles, but showed notable variation, suggesting that commuting distance could influence employee decisions regarding attrition. The average JobLevel was around level 2, with clear variation suggesting diverse levels of seniority and job responsibilities within the organization. YearsAtCompany averaged approximately 7 years, indicating a mix of newer and longer-tenured employees, while TotalWorkingYears averaged around 11 years, signifying considerable professional experience across the employee base. These varied numeric features collectively indicate their potential influence on employee attrition and underline their importance in understanding employee retention dynamics.

To select relevant features systematically, three methods were employed: Pearson Correlation, Mutual Information, and Random Forest feature importance. Pearson Correlation measures the linear relationship between numeric features and attrition, clearly visualized through a heatmap. Mutual Information assesses both linear and non-linear statistical dependencies between each feature and attrition, making it effective for capturing complex relationships. Random Forest feature importance calculates how significantly each feature contributes to reducing impurity within the decision trees, highlighting influential features regardless of linearity. Pearson correlation identified linear relationships, highlighting Age, MonthlyIncome, and TotalWorkingYears as strongly correlated with attrition (Figure 2). Mutual Information scores, which detect linear and non-linear dependencies, highlighted MonthlyIncome, TotalWorkingYears, StockOptionLevel, and OverTime as top predictors (Figure 3). Random Forest importance, measuring feature impact across tree ensembles, similarly emphasized OverTime, StockOptionLevel, Age, and YearsWithCurrManager among the strongest influences (Figure 4). Initially, the intent was to select only the top 10 features most strongly correlated with attrition to simplify the model and avoid potential overfitting. However, further analysis revealed that the predictive models did not exhibit signs of overfitting when utilizing all available features. Therefore, the decision was made to retain the entire feature set, recognizing that even those not strongly correlated features still provided valuable insights into the overall factors influencing employee attrition, enhancing the broader analytical understanding of attrition drivers.

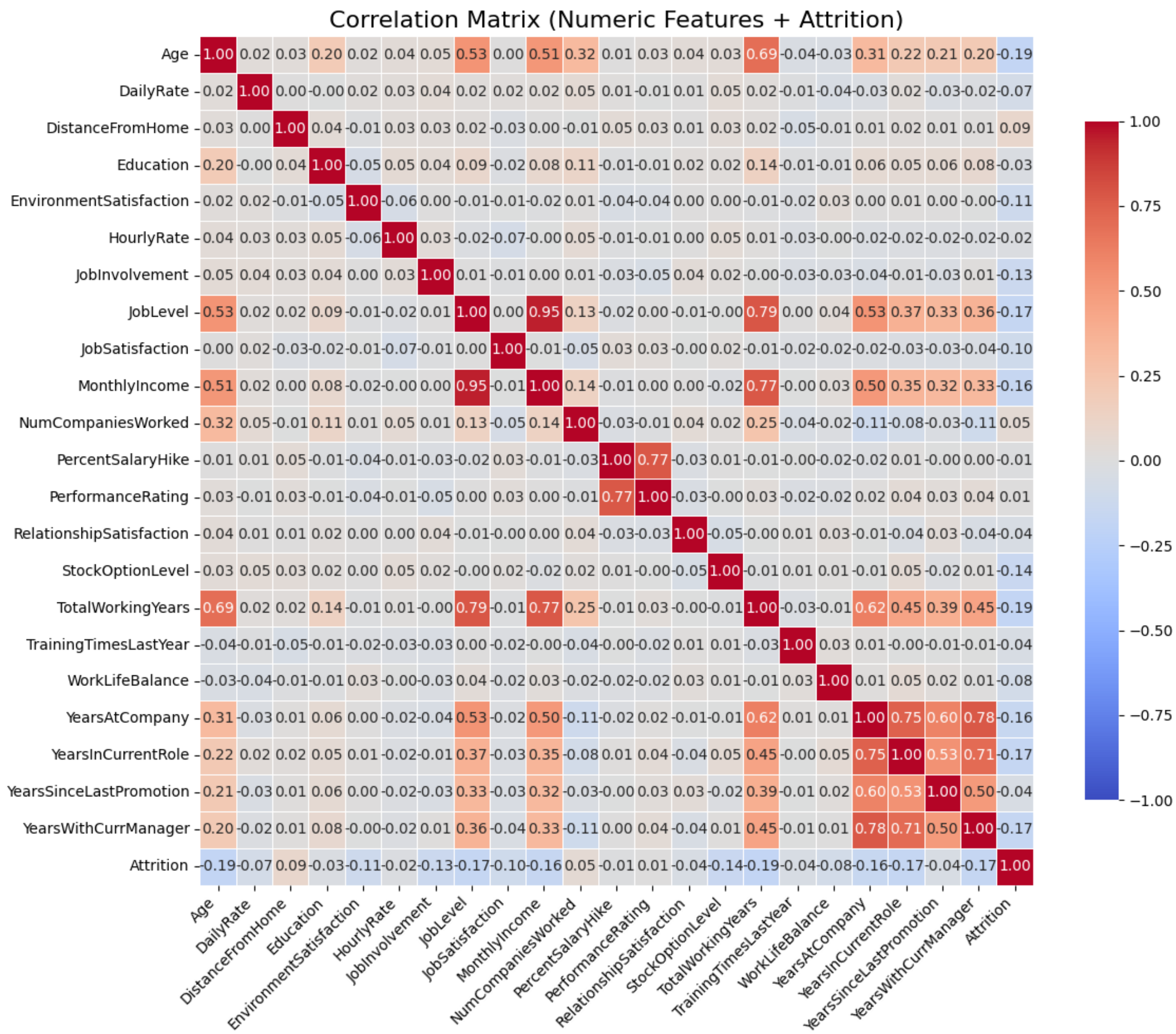

Figure 2: *Correlation Matrix Heatmap of Numeric Features and Employee Attrition*

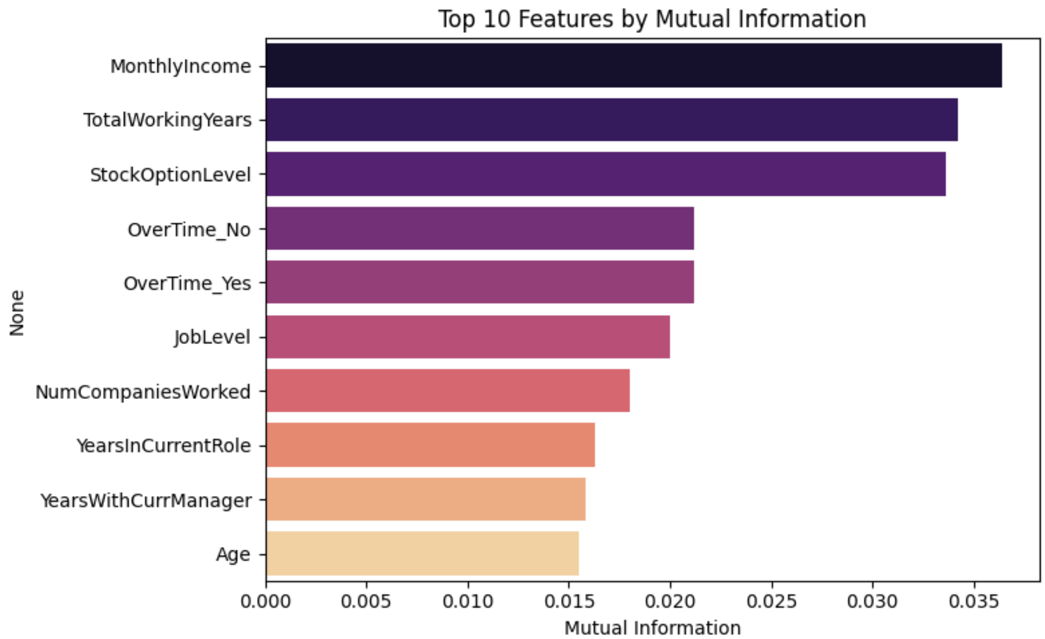

Figure 3: *Mutual Information Analysis: Top 10 Features Influencing Employee Attrition*

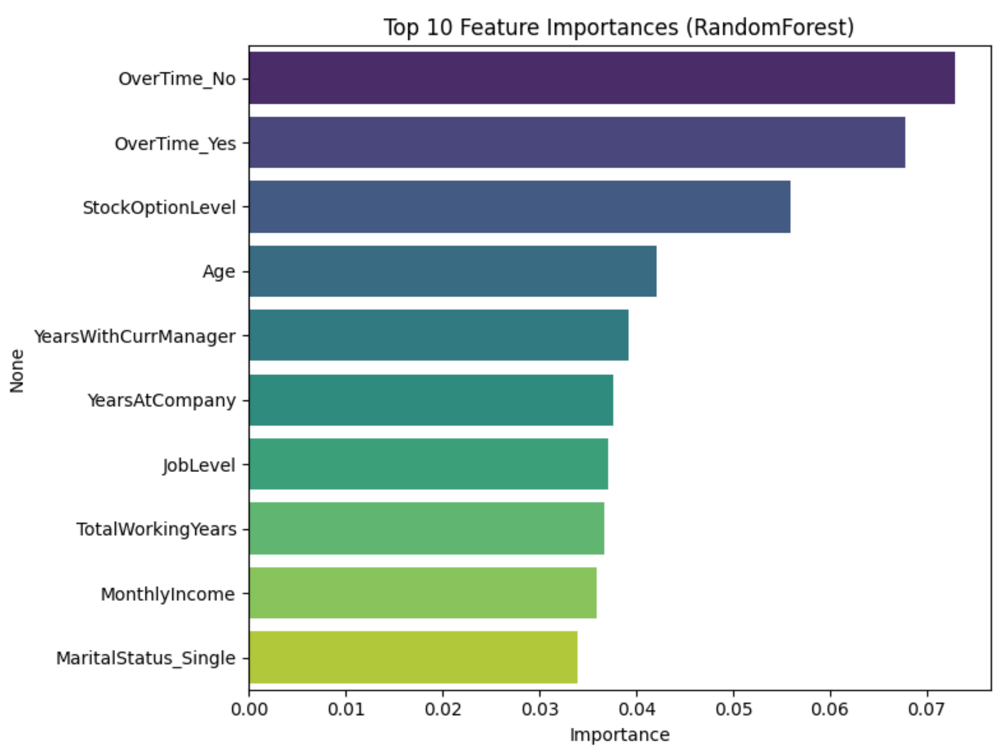

Figure 4: *Random Forest Feature Importance: Top 10 Predictors of Employee Attrition*

Given the high imbalance of attrition cases, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training dataset to balance class distributions without deleting any real-world data. SMOTE generates additional synthetic samples of employees who left the company by interpolating between existing minority examples, ensuring that predictive models can better recognize patterns associated with attrition (Figure 5). Importantly, SMOTE was applied exclusively to the training data, maintaining the realistic class distribution in the test data for accurate evaluation.

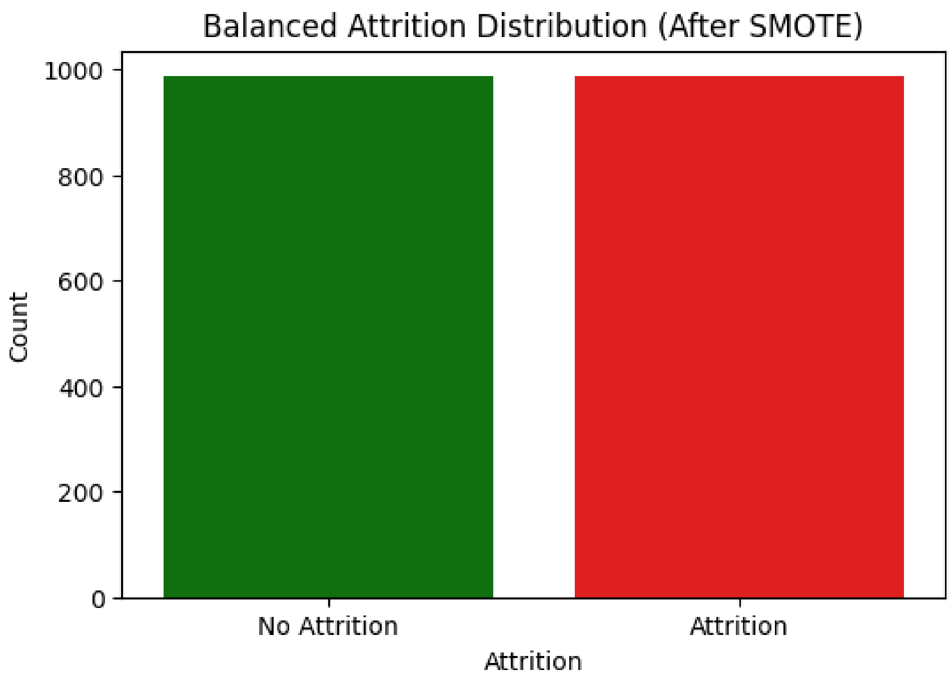

Figure 5: *Distribution of Employee Attrition After Applying SMOTE (Balanced Training Dataset)*

Additionally, all numeric features underwent standardization (feature scaling) to ensure consistent feature scales, supporting more stable and reliable model convergence and enhancing interpretability. Collectively, these preprocessing steps ensured robust and meaningful model development and subsequent analyses.

4.2 Model Development & Cross-Validation Results

This section outlines the development process of three supervised learning models: XGBoost, LightGBM, and a deep learning-based Multi-Layer Perceptron (MLP), all designed to predict employee attrition. XGBoost and LightGBM are both gradient boosting algorithms that build decision tree ensembles in a sequential manner, optimizing performance through advanced boosting techniques. XGBoost is particularly known for its regularization capabilities and speed, while LightGBM is optimized for efficiency and scalability, especially on large datasets. The Multi-Layer Perceptron (MLP), in contrast, is a type of deep neural network composed of multiple layers of interconnected nodes, suitable for capturing complex non-linear relationships but generally requiring larger datasets to perform effectively. The goal is to construct and train models using appropriate algorithmic pipelines and cross-validation techniques to ensure their robustness and suitability for further evaluation.

The XGBoost model was selected for its effectiveness in handling tabular data with high dimensionality. The model was trained on all available features after confirming that overfitting was not a concern. During development, 5-fold stratified cross-validation was employed to validate performance stability, resulting in a mean ROC AUC of approximately 0.792. In this validation approach, the dataset is divided into five equal parts (folds), ensuring that each fold preserves the same proportion of the target classes. Each fold is used once as a validation set while the remaining four serve as the training set, allowing for a more reliable estimate of the model's performance. The ROC AUC (Receiver Operating Characteristic - Area Under the Curve) is a metric that measures a model’ s ability to distinguish between classes; an AUC of 0.792 indicates that the model correctly ranks a randomly chosen positive example higher than a negative one approximately 79.2% of the time (Figure 6). The similarity between the cross-validation and hold-out test scores suggests that the model generalizes well and maintains consistency without overfitting.

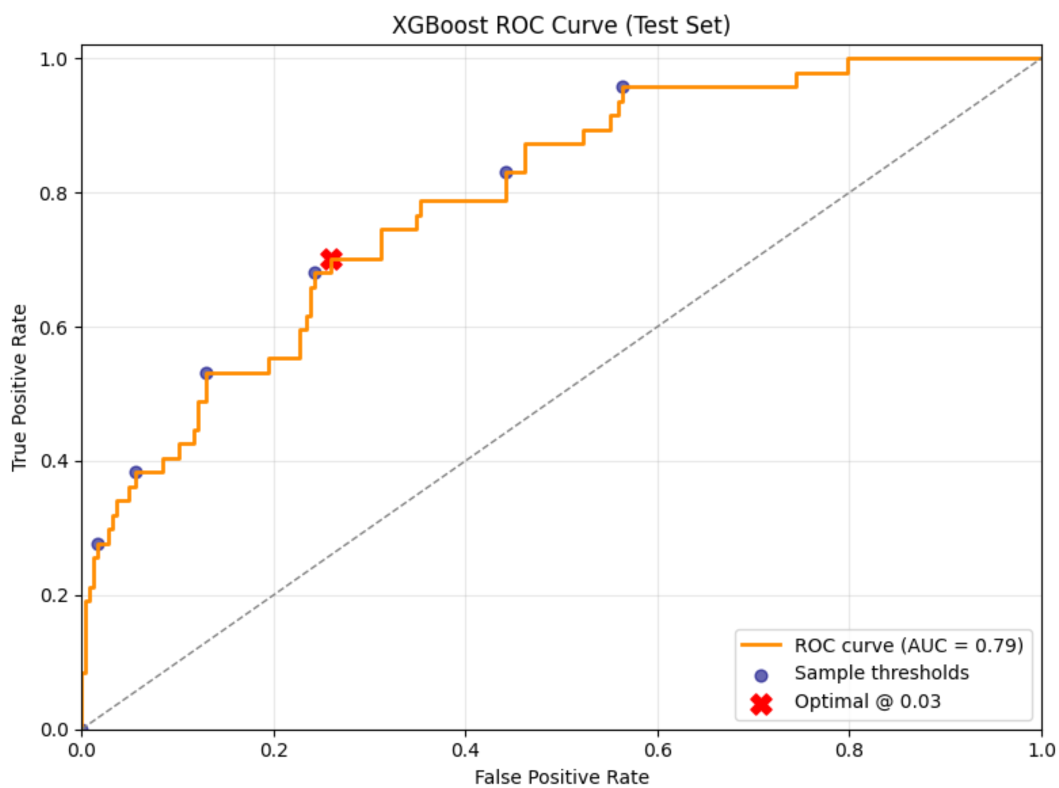

Figure 6: *ROC Curve of XGBoost Model on Test Set (AUC = 0.79)*

The LightGBM model produced the highest cross-validation score among the models tested. It achieved a mean ROC AUC of 0.815 with a relatively low standard deviation (±0.047), indicating strong performance and low variance across folds (Figure 7). The LightGBM pipeline demonstrated both computational efficiency and predictive strength, which ultimately led to its selection as the primary model for deployment in the later stages of the project.

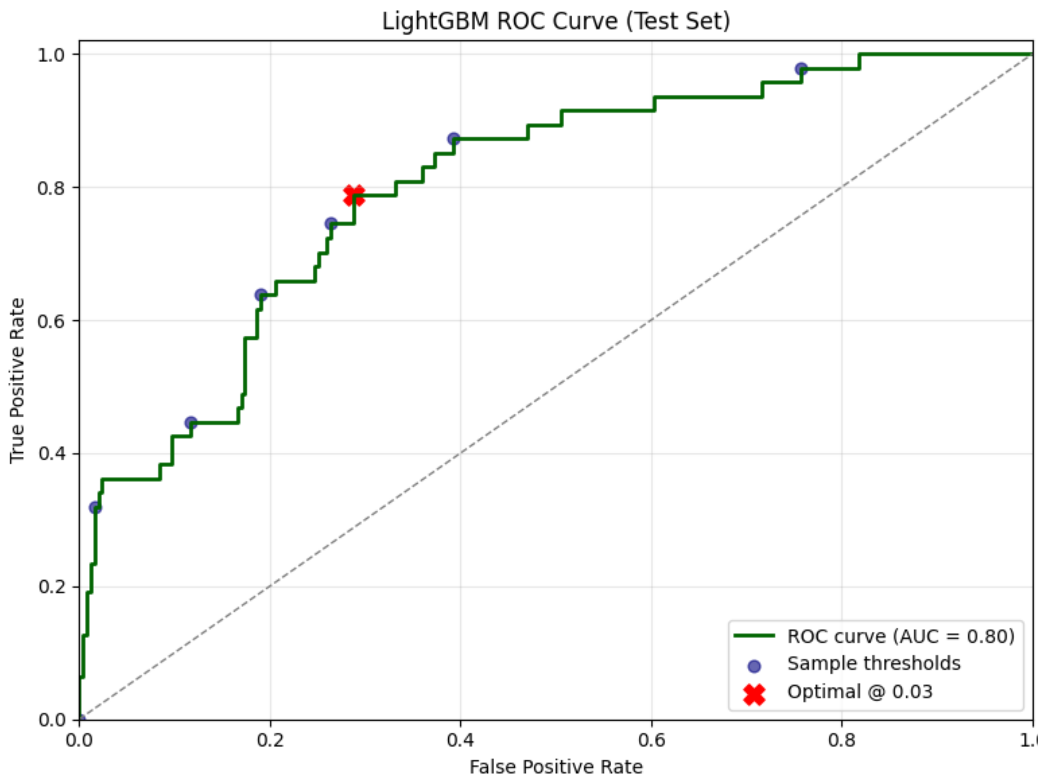

Figure 7: *ROC Curve of LightGBM Model on Test Set (AUC = 0.80)*

Additionally, a Multi-Layer Perceptron (MLP) model was explored as a deep learning alternative. This model consisted of three hidden layers and incorporated regularization techniques such as dropout and early stopping. Despite these efforts, the MLP demonstrated signs of overfitting and instability. While cross-validation results initially appeared promising, performance on the test set dropped significantly, with the AUC decreasing to approximately 0.68 (Figure 8). This outcome reflects the well-documented challenges of applying deep learning to small, structured datasets, where tree-based models typically outperform neural networks.

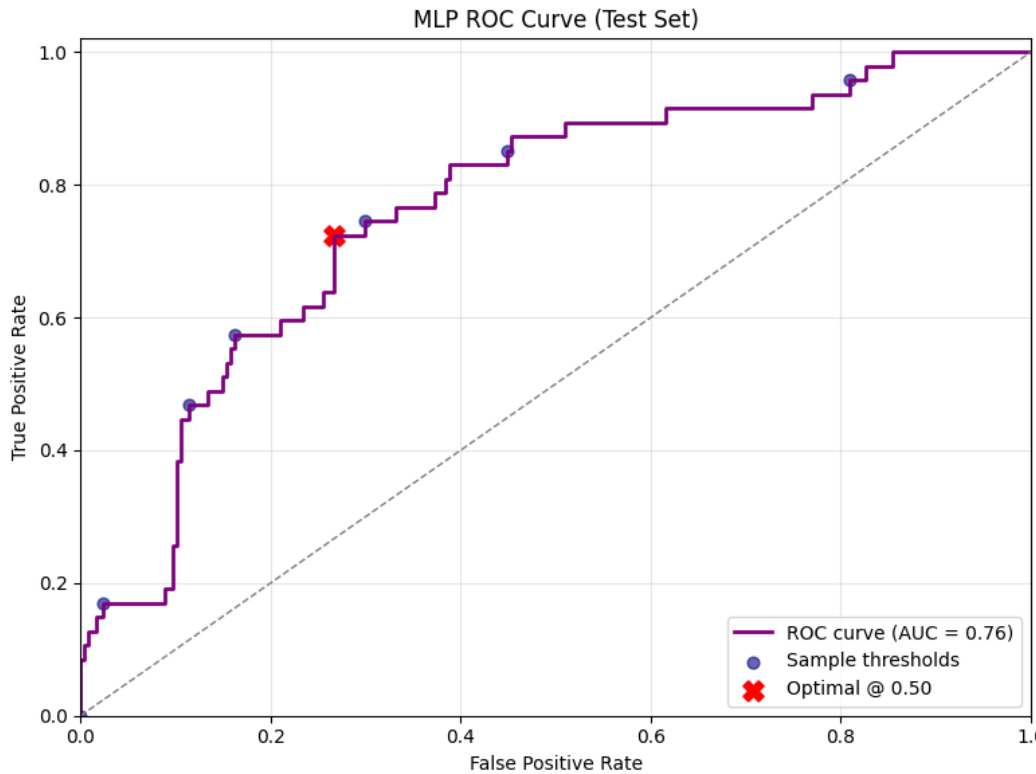

Figure 8: *ROC Curve of Multi-Layer Perceptron (MLP) Model on Test Set (AUC = 0.76)*

Overall, the model development process confirmed that tree-based ensembles, particularly LightGBM, are well-suited for structured HR datasets with a modest sample size. These models proceeded to the next phase of performance evaluation and interpretability analysis to support practical HR decision-making.

4.3 Hold-out Test Evaluation

Following model development and cross-validation, each pipeline was retrained on the complete training dataset and evaluated on a reserved hold-out test set consisting of 294 observations. A hold-out test set is a subset of the original dataset that is set aside during training and used exclusively to assess a model's performance on unseen data, thereby providing an unbiased estimate of its generalization ability. This section summarizes the classification performance of the three models-XGBoost, LightGBM, and Multi-Layer Perceptron (MLP)-on unseen data to assess their generalization capabilities (Table 1).

The XGBoost model achieved a test ROC AUC of 0.791, indicating solid discriminative performance. The classification report showed strong predictive ability for the majority class (no attrition), with precision of 0.884 and recall of 0.960. However, its ability to detect actual attrition cases was more limited, with a recall of 0.340 and an F1-score of 0.438 for the attrition class. These results suggest that while XGBoost is effective at identifying employees likely to stay, it has limitations in catching those likely to leave.

The LightGBM model outperformed the other models on most evaluation metrics. It achieved a test ROC AUC of 0.802, the highest among the three. Its recall for the no-attrition class was 0.984, and precision was 0.884. Most notably, it demonstrated improved precision for predicting attrition (0.789) compared to XGBoost (0.615), with a similar recall (0.319). The F1-score for the attrition class reached 0.455, surpassing the other two models. Additionally, the LightGBM model recorded the highest overall accuracy (0.878) and weighted average F1-score (0.855), reinforcing its suitability as the most balanced and generalizable model.

The MLP model, while included as an exploratory deep learning alternative, underperformed on the test set. Its ROC AUC was 0.638, significantly lower than both gradient boosting models. Although it showed reasonable recall for the attrition class (0.468), its precision was only 0.219, leading to a low F1-score of 0.348. This result confirms the challenges of applying deep neural networks to small, structured datasets, especially those with substantial class imbalance.

Model	**ROC AUC**	**Accuracy**	**Precision (Attrition)**	**Recall (Attrition)**	**F1-Score (Attrition)**	**Weighted F1-Score**
XGBoost	0.791	0.861	0.615	0.34	0.438	0.843
LightGBM	0.802	0.878	0.789	0.319	0.455	0.855
MLP	0.638	0.49	0.219	0.851	0.348	0.544

Table 1: *Models Evaluation Summary*

Overall, LightGBM emerged as the most effective model for predicting employee attrition on unseen data. It provided the best balance between precision and recall for the minority class while maintaining high performance on the majority class. Despite none of the models fully resolving the class imbalance challenge -most notably in recall for the attrition class- LightGBM’s superior metrics and generalization make it the most appropriate choice for further interpretability and fairness analysis.

In the following section, interpretability techniques such as SHAP will be applied to the LightGBM model to uncover feature-level insights, followed by an examination of fairness using the AI Fairness 360 toolkit.

4.4 Explainability and Model Understanding

To enhance model transparency and stakeholder trust, explainability techniques were applied to the best-performing classifier: LightGBM. The focus was on understanding which features most influenced attrition predictions and ensuring the model did not exhibit gender bias in its outputs.

SHAP (SHapley Additive exPlanations) values were used to interpret feature-level contributions. SHAP is a game-theoretic approach that quantifies how much each input feature contributes to a model's prediction, both in terms of direction (increasing or decreasing predicted attrition) and magnitude. It provides a consistent and model-agnostic framework for explaining predictions across individual employees.

Figure 9 (SHAP pipeline visualization) shows the full processing pipeline for the LightGBM classifier, including one-hot encoding, standardization, SMOTE rebalancing, and model fitting. The pipeline begins with a preprocessing stage in which categorical features are transformed into numerical format using one-hot encoding, a method that creates binary variables for each category level. This transformation ensures that machine learning models can interpret the categorical variables correctly without assuming any ordinal relationships. Next, standardization is applied to continuous numerical features through the StandardScaler, which rescales values to have a mean of zero and a standard deviation of one. This step is particularly important for algorithms that are sensitive to the scale of input data.

To address the class imbalance present in the dataset, SMOTE (Synthetic Minority Over-sampling Technique) is then applied. This technique generates new, synthetic examples of the minority class by interpolating between existing minority instances, thus ensuring that the model receives balanced exposure to both attrition and non-attrition cases during training. Finally, the processed data is passed to the LightGBM classifier, a high-performance gradient boosting framework optimized for speed and accuracy on structured data. The pipeline integrates all these steps into a streamlined and reproducible workflow that enhances both predictive performance and interpretability.

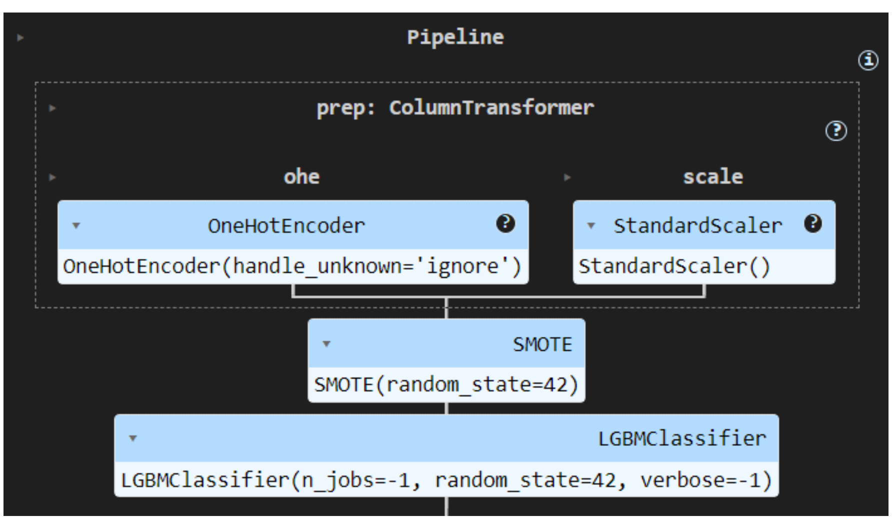

Figure 9: *SHAP pipeline visualization for the LightGBM classifier*

Figure 10 (SHAP summary plot) displays the average SHAP values across features. In this plot, each dot represents a single employee, with color indicating the feature value (red for high, blue for low) and horizontal position showing how strongly that feature influenced the model’s prediction pushing it toward higher or lower attrition risk. One of the most important predictors identified was OverTime_No. Employees who do not work overtime showed significantly lower predicted attrition risk, while working overtime was associated with a higher likelihood of leaving the organization. This supports existing HR intuition and suggests a key lever for intervention.

StockOptionLevel and EnvironmentSatisfaction also appeared prominently, with higher stock-option levels and greater satisfaction in the work environment linked to reduced attrition probabilities. Economic and demographic factors such as MonthlyIncome and Age showed a similar trend, as higher salaries and older age were generally predictive of retention. Additionally, managerial stability and job-related attitudes were influential: more years spent with the current manager and higher job satisfaction both contributed negatively to the predicted probability of attrition, reinforcing the value of managerial continuity and employee engagement. Toward the lower end of the importance spectrum, features such as TotalWorkingYears, YearsSinceLastPromotion, and specific job roles or departments carried some signal, but with less pronounced impact on predictions. Despite not being used to filter features, the SHAP analysis effectively confirmed and quantified the relevance of these variables, supporting both model transparency and actionable business insight.

These insights provide clear guidance for human resource professionals. For example, organizations may consider reducing excessive overtime, enhancing stock compensation plans, and investing in workplace satisfaction initiatives to improve retention outcomes.

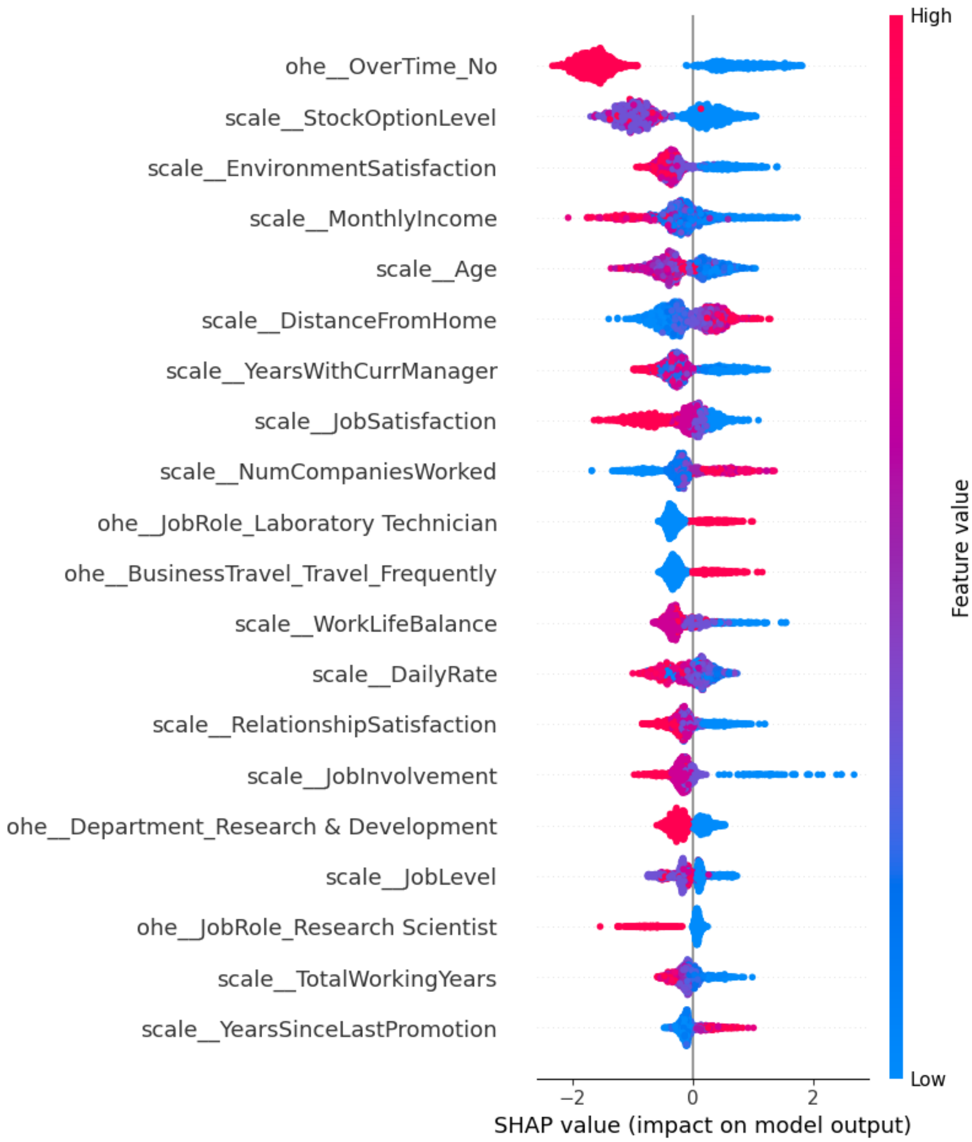

Figure 10: *SHAP summary plot*

In addition to interpretability, fairness was assessed using IBM’s AI Fairness 360 toolkit, with gender chosen as the protected attribute. Two fairness metrics were computed: Disparate Impact and Statistical Parity Difference. The Disparate Impact value of 1.008 indicated near-perfect parity in predictions between male and female employees, suggesting that both groups were treated equitably by the model. The Statistical Parity Difference was approximately +0.007, indicating that women received slightly more favorable outcomes than men, although by a negligible margin that falls well within commonly accepted thresholds for fairness auditing.

In summary, the SHAP analysis revealed clear, interpretable relationships between several important features and attrition risk, providing human resource teams with concrete areas of focus. Moreover, fairness diagnostics confirmed the model’s equitable behavior with respect to gender, reinforcing its reliability and ethical suitability for deployment in a real-world HR analytics context.

4.5 Deployment

Finally, the best performing developed predictive model is prepared for practical deployment to demonstrate how the model could be leveraged by HR practitioners to obtain actionable insights to implement targeted retention strategies effectively. This process includes serializing the model using a python pickle file that can be uploaded to a GCP function. This GCP function is responsible for loading the model, receiving HTTP POST requests with JSON payloads containing the required features for running the model, wrapping the features into a pandas dataframe, running the model on the provided features, then finally providing the attrition probability and attrition label as a JSON response. This method of using a GCP function with a REST endpoint allows for seamless integration into various types of applications. To demonstrate practical business value, a proof-of-concept React web application was developed (Figure 11). This web application allows users to upload CSV files containing independent variables for employee attrition. The web application processes each row by making calls to the GCP function endpoint, which returns both the attrition probability and attrition label. The results for each row are then appended to the original CSV file and newly modified CSV containing both the original features as well as the results are then made available to be downloaded by users, showcasing the operational capability and real-world applicability of the predictive model to handle large amounts of data and provide real world actionable insights to HR departments.

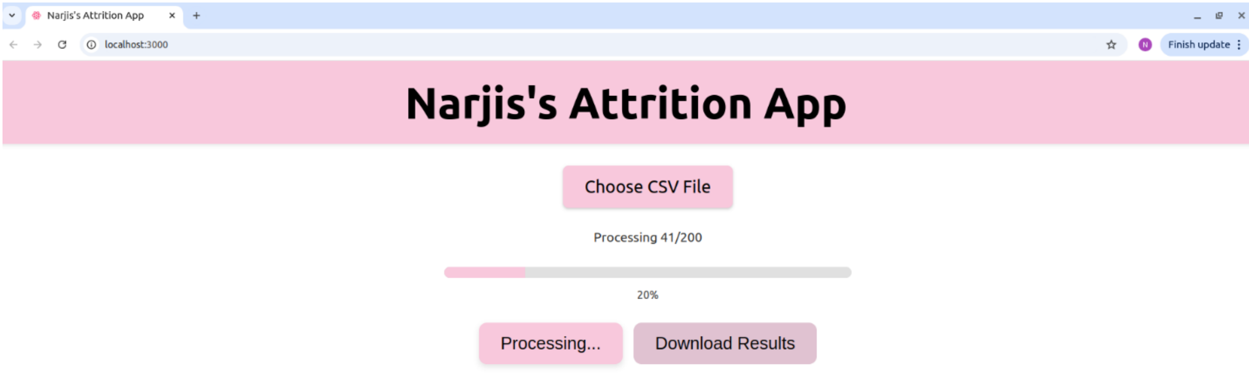

Figure 11: *Attrition Web App Front End (Developed using React) processing rows of a CSV file*

An important note for deployment of ML models to GCP functions is the necessity of matching dependency versions. Special care was taken to ensure that the provided requirements.txt file that was uploaded to GCP contained all the same package versions that were used in the training and evaluation of the model to ensure seamless transfer to the cloud and to reduce the risk of mismatched code versions.

Deploying the attrition prediction model as a GCP function provides an elegant, serverless solution that enables on-demand, scalable inference without incurring the overhead of infrastructure management. By encapsulating the serialized best-performing pipeline within a stateless HTTP endpoint, the approach leverages automatic horizontal scaling to accommodate variable request volumes—whether invoked interactively from an HR dashboard or orchestrated in batch via scheduled workflows—while ensuring cost-efficiency through fine-grained, pay-per-invocation billing. Furthermore, seamless integration with Google's logging, monitoring, and versioning tools streamlines validation and iterative improvements, allowing the research to focus on model refinement rather than operational constraints. This deployment paradigm thus elevates experimental reproducibility and accelerates the translation of predictive insights into actionable retention strategies.

5. Retention Strategies Discussion

The SHAP analysis in this study revealed multiple key features contributing to employee attrition, each offering actionable insight into potential intervention points. The most influential predictor OverTime emerged as a critical factor. Employees working overtime were significantly more likely to leave their job. This suggests that organizations should monitor and manage employee workloads, ensuring overtime is not excessively relied upon. HR departments can implement policies that limit mandatory overtime, promote flexible work schedules, and encourage a better work-life balance.

The model also identified StockOptionLevel and MonthlyIncome as important economic factors influencing attrition. Employees with higher stock options and salaries were less likely to leave. To address this, firms may consider revisiting compensation structures—not only through direct pay increases but also through equity participation or long-term incentive plans that foster a sense of ownership and long-term commitment among employees.

EnvironmentSatisfaction and JobSatisfaction were also among the top predictors, indicating that non-monetary workplace factors play a substantial role in retention. Employers should therefore invest in workplace climate and culture, ensuring employees feel valued, supported, and aligned with the organization's values. Regular engagement surveys, wellness programs, and employee recognition initiatives can help foster a more satisfying work environment.

Age and YearsWithCurrManager also surfaced as strong predictors. Younger employees and those with shorter tenures under a manager were more likely to leave. This highlights the value of targeted onboarding programs, mentorship, and leadership development, especially for new employees or those reporting to newer or less experienced managers. Enhancing management training to improve relationships with direct reports can also help mitigate attrition.

Lastly, even features with smaller individual impacts—such as YearsSinceLastPromotion and TotalWorkingYears—point to the importance of career growth and internal mobility. Employees who do not perceive clear advancement opportunities are more prone to disengagement and departure. HR departments should prioritize transparent promotion pathways and upskilling programs to retain high-potential talent.

Taken together, these findings advocate for a holistic retention strategy that combines fair compensation, balanced workloads, workplace satisfaction, strong manager relationships, and career progression. By aligning retention initiatives with these evidence-based drivers, organizations can design more effective and targeted interventions to reduce voluntary turnover and foster long-term employee engagement.

6. Predicting Employee Performance Discussion

The final phase of this research project aimed to explore whether machine learning techniques could be applied to predict employee performance using the same dataset previously utilized for attrition modeling.

Specifically, the target variable examined was PerformanceRating, a categorical feature within the IBM HR dataset. The rationale for maintaining a consistent dataset throughout this project was to allow for methodological continuity and to support a unified data-driven approach across attrition, retention strategy, and performance analysis.

Initial efforts focused on replicating the modeling pipeline established during the attrition prediction phase, applying both XGBoost and LightGBM classifiers. However, the results were consistently unsatisfactory. Both models exhibited significant performance degradation, with test ROC AUC scores of 0.541 and 0.565 respectively, barely exceeding random-chance prediction. While the classifiers achieved high precision and recall for the majority class (employees rated "excellent"), they were largely ineffective at identifying the minority class (those rated "outstanding"). This imbalance was reflected in the low F1-scores and poor recall metrics for the target class.

Upon further investigation, it became evident that the dataset itself posed inherent challenges. The PerformanceRating variable was highly imbalanced, containing only two classes—"excellent" and "outstanding"—with a disproportionately small number of observations in the latter category. Furthermore, during the initial modeling phase, it was discovered that the feature PercentSalaryHike had a strong, potentially data-leaking correlation with the target variable. Once this feature was removed, model performance dropped substantially, further underscoring the fragility of predictive signals in this domain.

These findings highlight a broader issue: employee performance ratings are often subjective, sparsely distributed, and shaped by numerous contextual, organizational, and interpersonal factors that are not captured in structured HR datasets. Performance assessments tend to exclude low performers in publicly available datasets due to privacy, ethical, and reputational concerns. This selective visibility introduces bias and severely limits the generalizability of predictive models.

From an applied perspective, the inability to effectively model performance reinforces the need for organizations to treat performance management as a multidimensional and often qualitative process. Employee performance is influenced by a variety of dynamic and context-specific factors—including individual motivation, team dynamics, leadership quality, organizational culture, and external life circumstances—that are difficult to quantify or capture in static HR records. Performance also evolves over time and may vary depending on the nature of the work, feedback received, and opportunities for growth. Thus, rather than relying solely on predictive tools, companies should adopt a more holistic approach to evaluating and supporting performance. This includes establishing continuous feedback mechanisms that encourage two-way communication between employees and managers, implementing tailored development plans that align with individual strengths and career aspirations, and conducting robust performance appraisals that combine both quantitative indicators (e.g., project outcomes, KPIs) and qualitative inputs (e.g., peer feedback, behavioral assessments, and leadership evaluations). Integrating these diverse perspectives enables a more accurate and fair assessment of employee contributions, and better supports long-term growth, engagement, and retention.

7. Contributions, Implications & Limitations

This paper contributes to the field of people analytics by developing, evaluating, and interpreting machine learning models to predict employee attrition, analyze drivers of retention, and explore the feasibility of predicting employee performance. One of the core contributions lies in the successful implementation of a robust LightGBM-based pipeline that integrates SMOTE, SHAP-based explainability, and fairness diagnostics using AI Fairness 360. This approach not only achieved high predictive performance on attrition outcomes but also offered interpretable and equitable predictions—demonstrating how advanced machine learning techniques can support human resource decision-making in a responsible and transparent manner.

Practically, the findings from the attrition modeling and SHAP analysis provide data-driven guidance to HR practitioners seeking to reduce turnover. Insights related to overtime, compensation, satisfaction, and managerial stability can inform targeted interventions that align with employee needs and organizational goals. The interpretability framework further enables stakeholders to trust and act on the model's predictions, reinforcing the real-world utility of explainable AI in workforce analytics.

The second major implication of this project lies in its emphasis on the challenges of performance prediction. The exploration of employee performance prediction revealed fundamental limitations in applying machine learning to highly subjective and imbalanced performance ratings. This work underscores the need for organizations to approach performance management with a broader, multidimensional framework that incorporates qualitative assessments, behavioral evaluations, and continuous feedback loops.

In future research, more comprehensive datasets that include diverse performance bands and incorporate behavioral, task-level, and 360-degree feedback could provide a more viable foundation for predictive modeling. Until such datasets are made available, efforts to model performance using limited, static features are likely to remain constrained by both technical and ethical boundaries.

Despite its contributions, this study is not without limitations. First, the dataset used is publicly available and lacks demographic diversity and organizational context, which may limit the generalizability of the results. Additionally, while SMOTE was employed to rebalance classes in the attrition task, such synthetic techniques may not fully capture the complexity of real-world employee behavior. Furthermore, fairness was assessed only along the gender axis; future work should explore additional protected attributes such as race, age, and disability status.

Overall, this thesis illustrates the potential—and boundaries—of using machine learning for strategic HR analytics. It highlights the importance of combining predictive accuracy with interpretability, ethical scrutiny, and domain-specific understanding to inform more effective and human-centered workforce decisions.

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